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AIM:TIME (AIM:TIME) SERIES ANALYSIS USING R DATASET:AIMILES,On this R-data statistics page, you will find information about the airmiles data set which pertains to Passenger Miles on Commercial US Airlines, 1937–1960. The airmiles data set is found in the datasets R package

```
## Load the Forecast Package into RStudio
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method           from
##   as.zoo.data.frame zoo
```

```
## Load the iris Dataset and View Its Class
data("airmiles")
class(airmiles)
```

```
## [1] "ts"
```

```
## Display
airmiles
```

```
## Time Series:
## Start = 1937
## End = 1960
## Frequency = 1
## [1] 412 480 683 1052 1385 1418 1634 2178 3362 5948 6109 5981
## [13] 6753 8003 10566 12528 14760 16769 19819 22362 25340 25343 29269 30514
```

```
## check on our date values
start(airmiles)
```

```
## [1] 1937 1
```

```
end(airmiles)
```

```
## [1] 1960 1
```

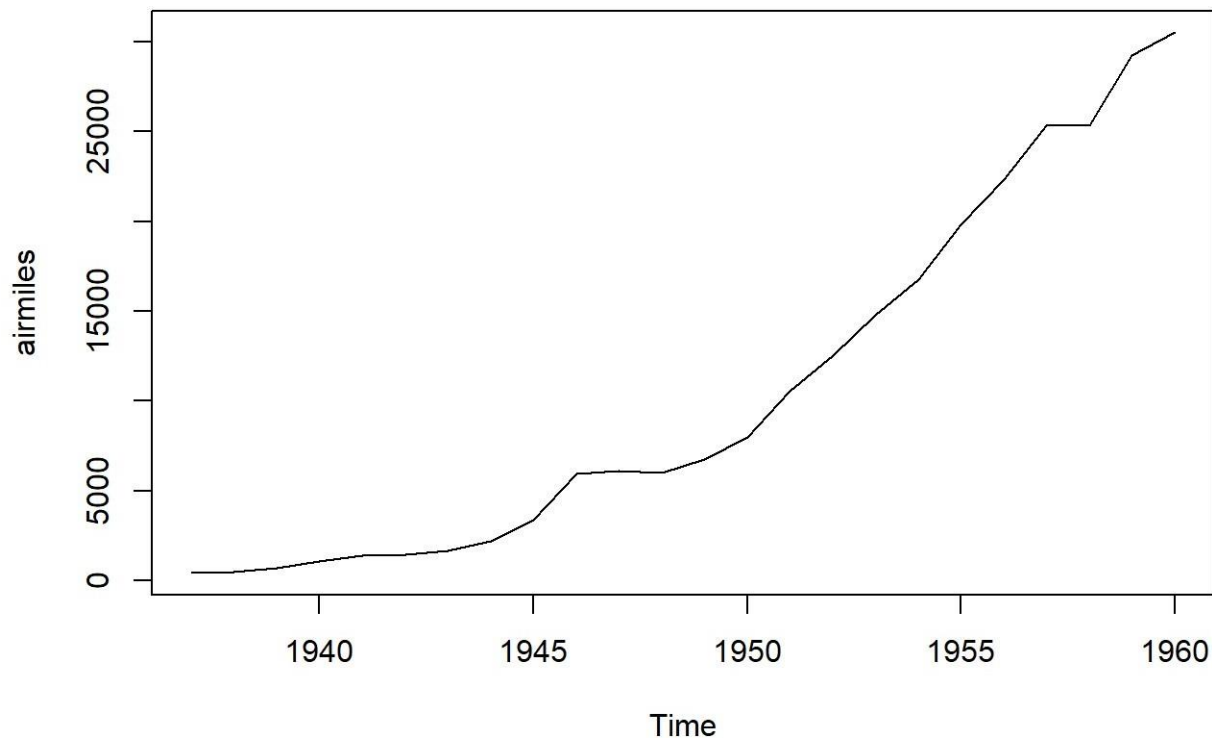
```
# Find out If There Are Any Missing Values  
sum(is.na(airmiles))
```

```
## [1] 0
```

```
## Check the Summary of the Dataset  
summary(airmiles)
```

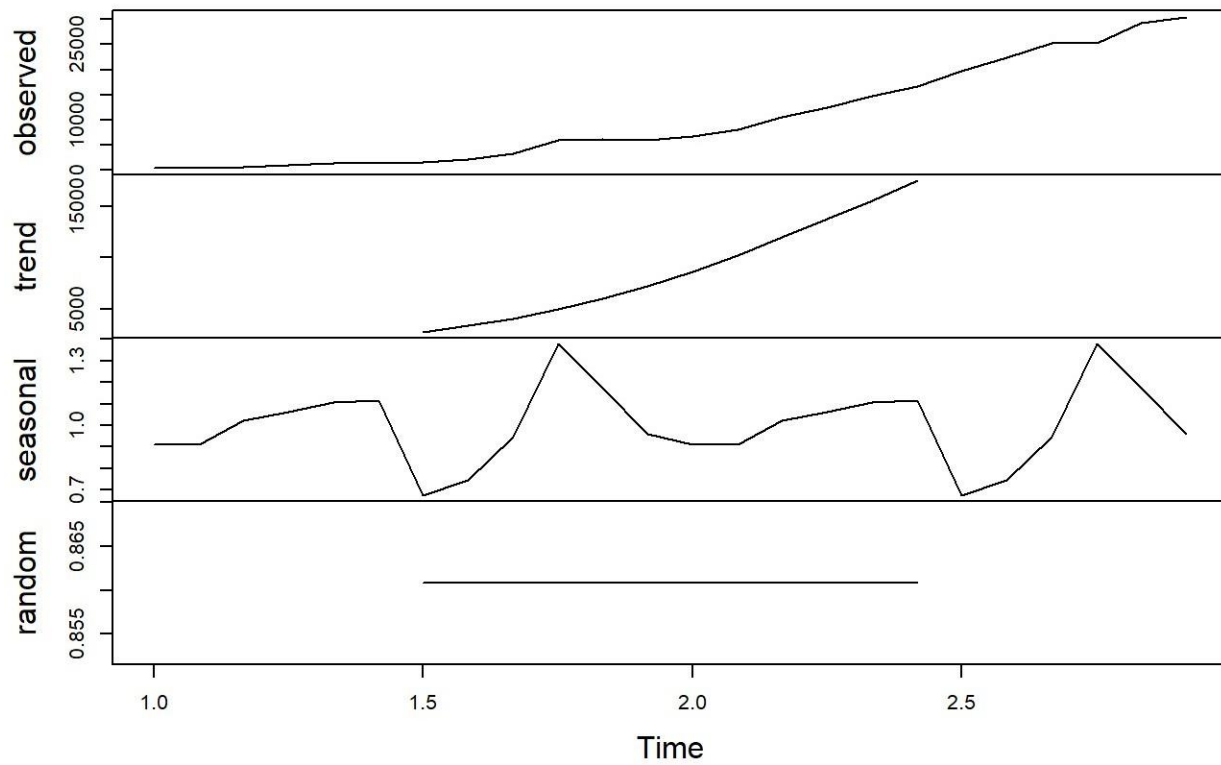
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##      412   1580   6431   10528   17532   30514
```

```
##Plot the Dataset  
plot(airmiles)
```

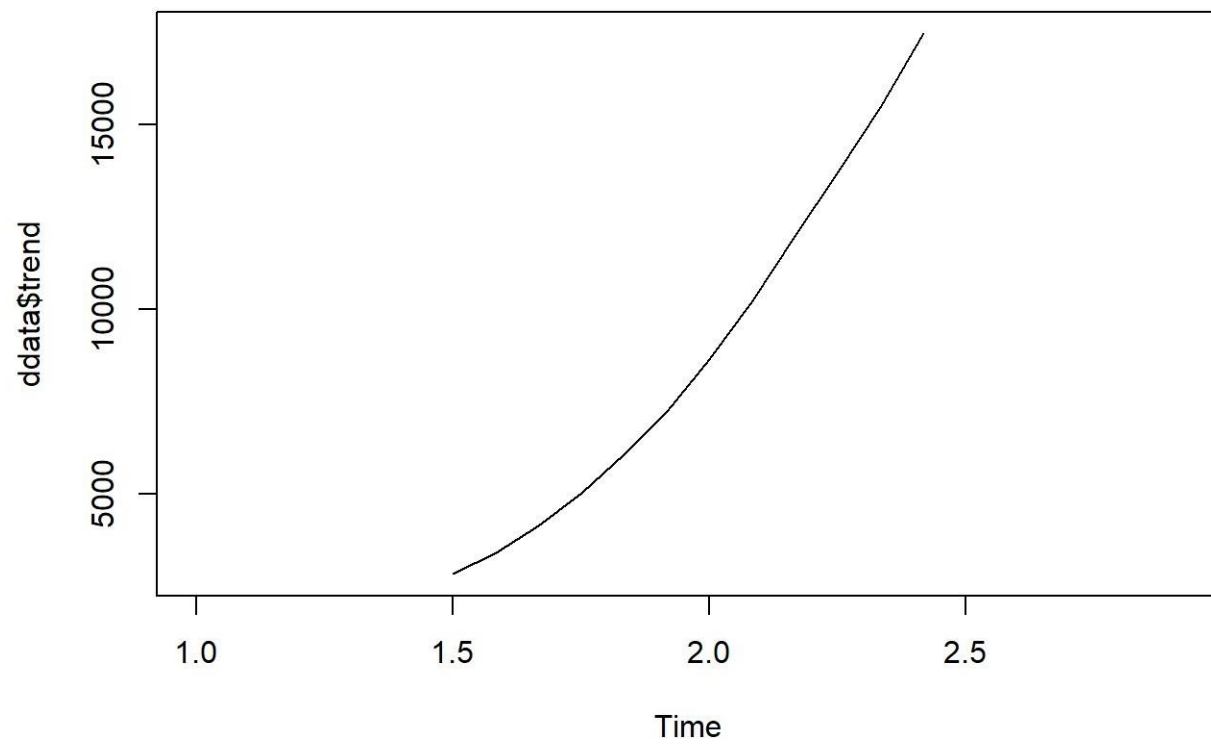


```
## Decompose the Data Into Four Components  
tsdata <- ts(airmiles, frequency = 12)  
ddata <- decompose(tsdata, "multiplicative")  
plot(ddata)
```

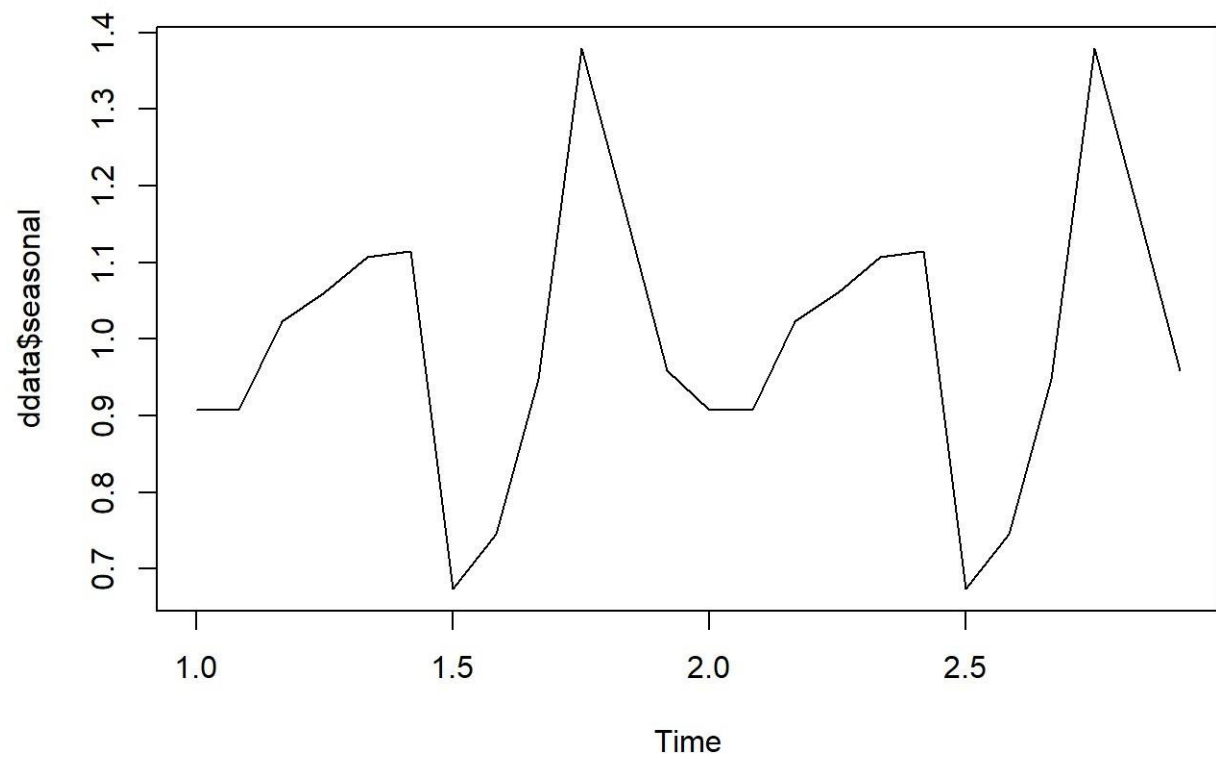
Decomposition of multiplicative time series



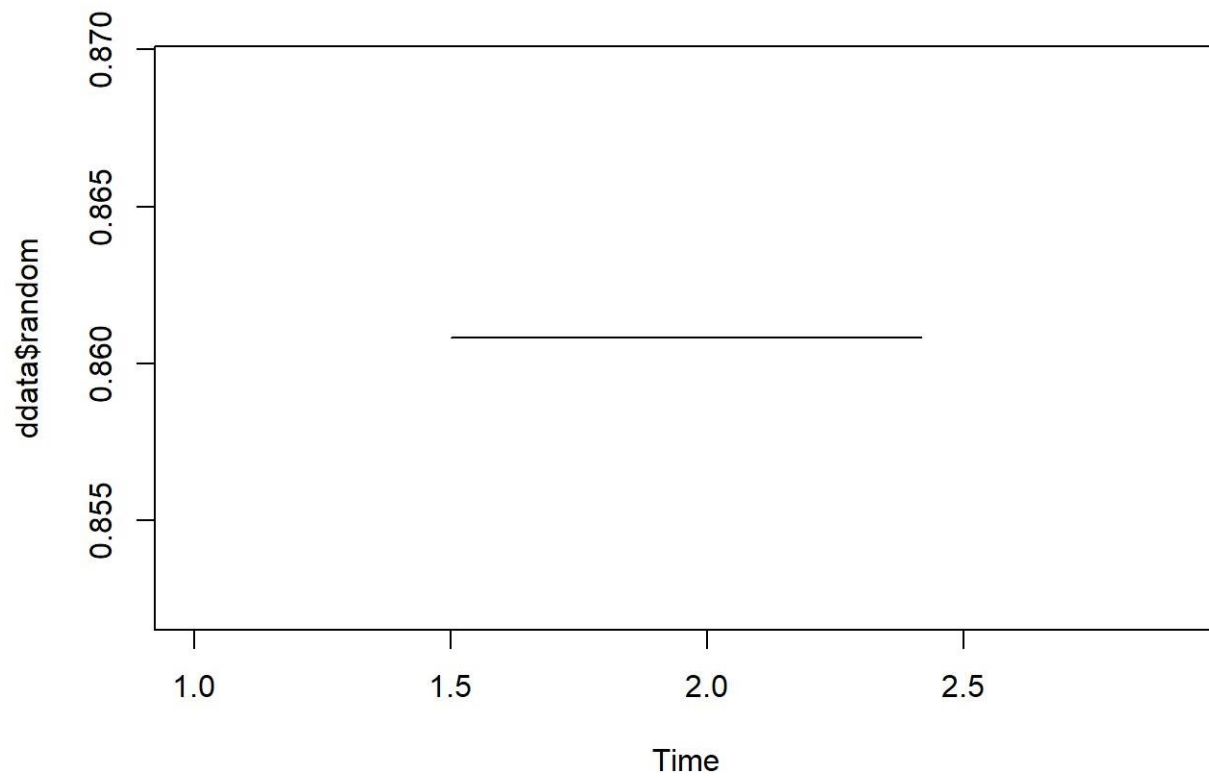
```
## Plot the Different Components Individually  
plot(ddata$trend)
```



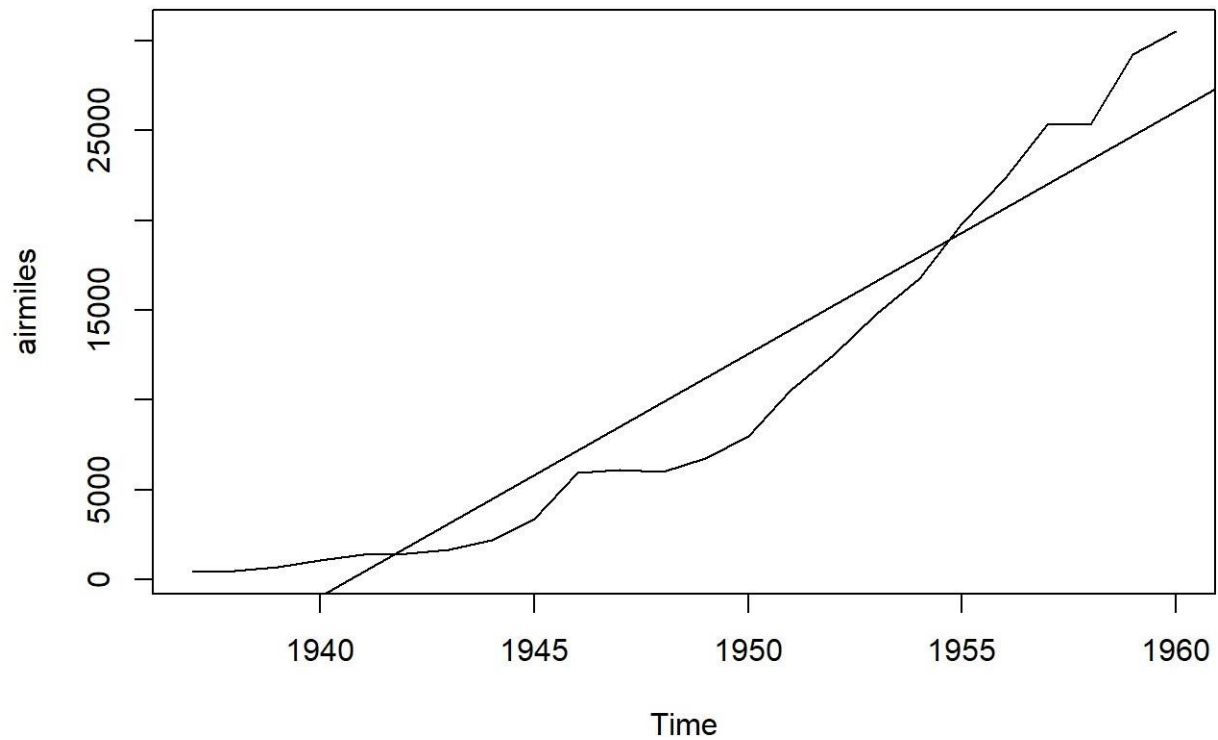
```
plot(ddata$seasonal)
```



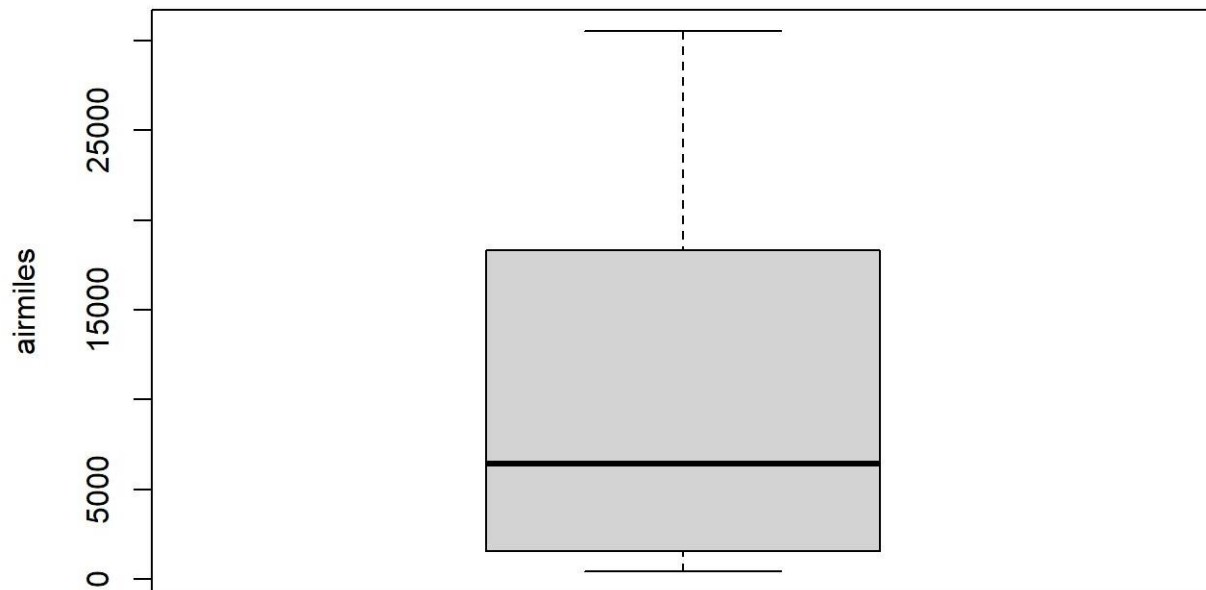
```
plot(ddata$random)
```



```
## Plot a Trendline on the Original Dataset  
plot(airmiles)  
abline(reg=lm(airmiles~time(airmiles)))
```



```
## Create a Box Plot by Cycle  
boxplot(airmiles~cycle(airmiles, xlab="Date", ylab = "Passenger Numbers  
(1000's)", main = "Monthly air passengers boxplot from 1949-1960"))
```



```
s, xlab = "Date", ylab = "Passenger Numbers\n(1000's)", main = "Monthly air passengers boxplot
```

```
## Build the ARIMA Model Using auto.arima() Function
mymodel <-auto.arima(airmiles)
mymodel
```

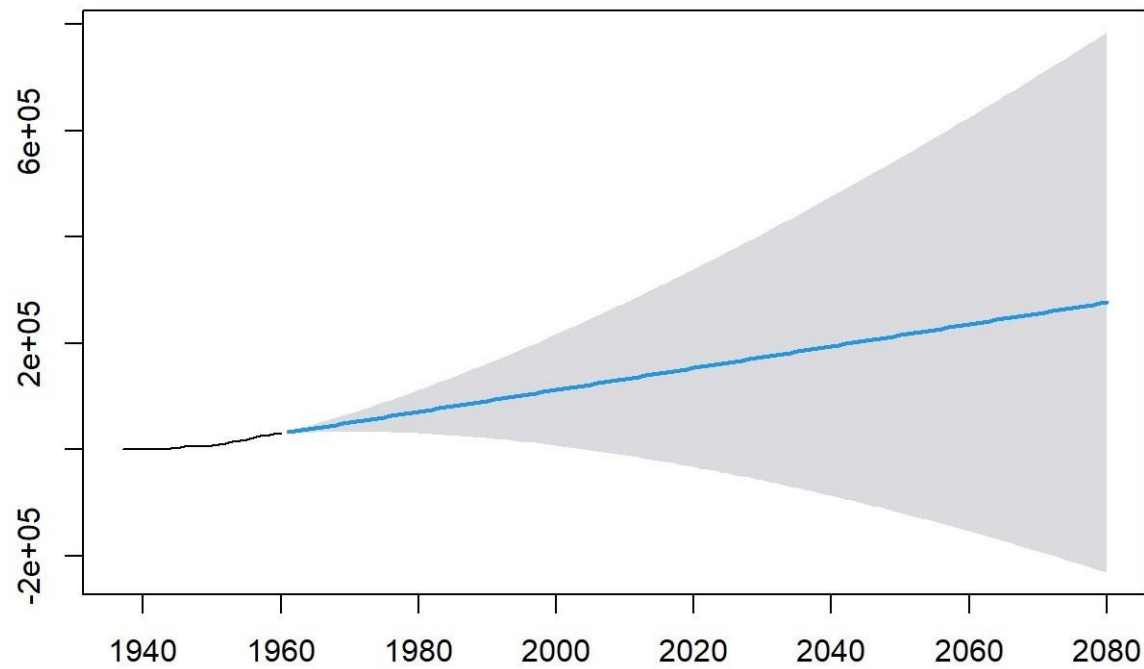
```
## Series: airmiles
## ARIMA(0,2,1)
##
## Coefficients:
##          ma1
##        -0.7031
## s.e.    0.1273
##
## sigma^2 = 1234546: log likelihood = -185.33
## AIC=374.67   AICc=375.3   BIC=376.85
```

```
## Plot the Residuals
plot.ts(mymodel$residuals)
```




```
# #Forecast the Values for the Next 10 Years  
myforecast <- forecast(mymodel, level=c(95), h=10*12)  
plot(myforecast)
```

Forecasts from ARIMA(0,2,1)



```
Box.test(mymodel$resid, lag=5, type="Ljung-Box")
```

```
##  
## Box-Ljung test  
##  
## data: mymodel$resid  
## X-squared = 4.7529, df = 5, p-value = 0.4468
```

...